

Biodiversity in the 21st Century at Gbg University

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1 Reading 1: Diaz et al 2019

1.1 Summary

- **telecoupling** – human actions increasingly act at a distance due to globalization
- increasing demand between supply and demand due to global trade
- the analysis pinpoints five crucial levers (priority interventions) and eight leverage points (for intervention)

1.2 Intro

- the human impact on life on Earth has increased since the 1970s
- both the benefits of economy and the cost of reducing nature are unequally distributed

1.3 Taking stock of the fabric of life

- over the past 50 years, the quality of nature to support life has declined on 14 of 18 categories identified by IPBES ¹
- exceptions to the downward trend are: regulation of ocean acidification, energy, food and feed, materials and assistance
- more than 800 million people still face chronic food deprivation
- the biomass of world's vegetation has halved over human history
- forest area is only 68% of its preindustrial size

1.4 Direct and indirect drivers of change

- Direct drivers:
 - land/sea use change
 - direct exploitation
 - climate change
 - pollution
 - invasive alien species
 - others
- Indirect drivers:
 - demographic and sociocultural
 - economic and technological

- institutions and governance
- conflicts and epidemics

- examples of declines in nature:
 - natural ecosystems have declined by 47% on average, relative to their previous states
 - 25% of species are already threatened by extinction
 - biotic integrity – abundance of naturally present species
 - has declined by 23% on average in terrestrial communities
 - the global biomass of wild mammals has fallen by 82%
 - 72% of indicators developed by Indigenous Peoples and local communities have deteriorated

1.5 Progress towards internationally agreed goals

- 20 Aichi Targets in the Strategic Plan on Biodiversity 2011-2020
 - contains 54 elements
 - good progress has been made on 5
 - moderate progress towards 19
 - poor progress or movement away from target on 21
 - unknown progress on 17 elements
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1.6 Levers and leverage points for transformative change

Levers:

- incentives and capacity building
- cross-sectional cooperation
- pre-emptive action
- decision-making in the context of resilience and uncertainty
- environmental law and implementation

Leverage points:

- embrace diverse vision of a good life
- reduce total consumption and waste
- unleash values and action
- reduce inequalities

¹Intergovernmental Platform on Biodiversity and Ecosystem Services

- practice justice and inclusion in conservation
- internalize externalities and telecouplings
- ensure environmentally friendly technology, innovation and investment
- promote education and knowledge generation and sharing

2 Lecture 1

2.1 Housekeeping

- Every Tuesday
- 18 - ca. 20h
- Live or recorded on Zoom
- All lectures will be recorded and available on Canvas
- Two in-person meetings
- Gothenburg Natural History Museum
- Gothenburg Botanical Garden
- Assignment: develop and submit an essay on a topic we cover in the course that is of particular personal interest. Open-ended and offers the opportunity for personal reflection, while also gaining experience in more scientific-styled writing.

2.2 Biodiversity bias

- 2 million – 1 trillion species on Earth
- ca. 1.7 million databased
- Knowledge derives from charismatic, macroscopic species.

2.3 Mechanisms of biodiversity formation

- speciation
- extinction
- migration

Adaptation = the process which enables organisms to adjust to their environment in order to ensure survival. This can lead to speciation.

Example: polar bears have extremely large feet to distribute weight better on snow.

2.4 Speciation

Types of speciation:

- allopatric – populations get geographically distanced which limits the gene flow
- peripatric – a small group gets isolated
- parapatric – populations are adjacent but not completely separated
- sympatric – species evolve from a single ancestral species while living in the same area

2.5 Extinction

The termination of any organism.

Species have a lifespan, on average 2 million years. After that, species either speciate or go extinct.

2.6 Dispersal

Dispersal = movement of individuals (through geological time, such as marsupials colonizing North America once South and North Americas got connected through geological time)

Since the industrial revolution, many species are progressively changing their distribution and dispersing further north.

2.7 Latitudinal species gradient

Closer to equator/tropics = more species (also more human languages!)

Does the diversity in species and language covary?

2.8 Species

How do we measure species for example in tropical jungle with countless species?

Quadrants: count within a small area then you can extrapolate

Species richness = the number of different species in an area

Numerical species richness = number of species per specific number of individuals

Species density = number of species per unit area

Species abundance = the relative abundance of species

Amazonia is known to be highly biodiverse, but a few species are very abundant and most species are incredibly rare. 1% of species make up for 50% of all trees.

2.9 Learning outcomes

- Biodiversity: number of differential biological variants found in a given place and time
- Speciation, extinction, and dispersal build biodiversity
- Biodiversity is not evenly distributed and can be measured in different ways

3 Reading 2: Forest et al 2018: Gymnosperms on the EDGE

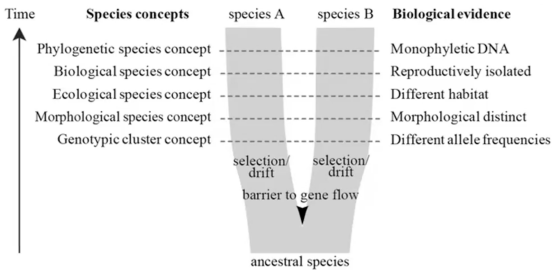
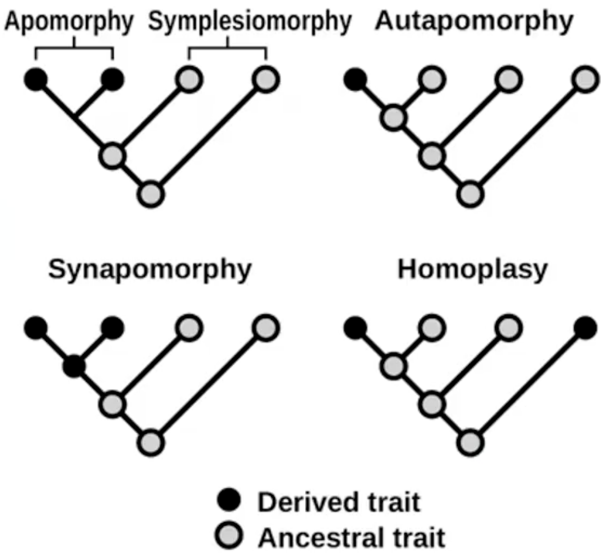
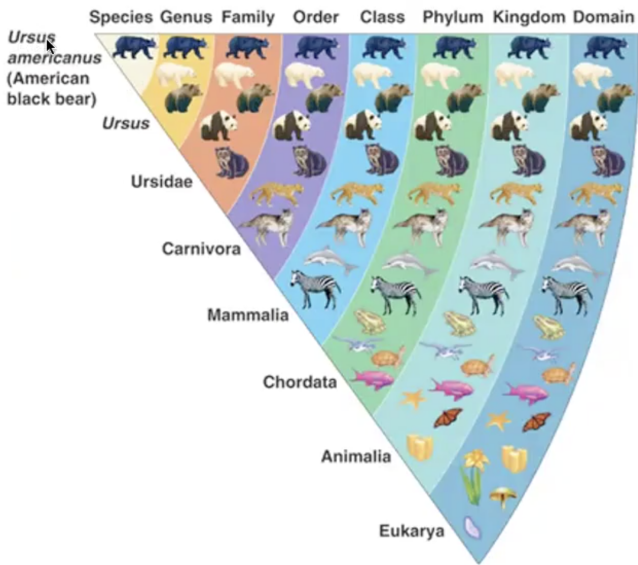
3.1 Gymnosperms



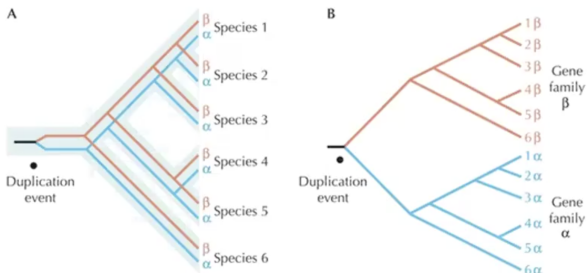
- Comprising ginkgo, conifers, cycads, and gnetophytes
- 40% species at high risk of extinction
- For plants in general, this number is 20%

3.2 Linneus’ binomial classification rules

1. The entire 2-part name must be written in italics.
2. The genus name must be written first.
3. The genus name must be capitalized.
4. The specific epithet is never capitalized.



Gene trees vs. species trees



4 Climate, Oxygen, and the Future of Marine Biodiversity

- The ocean enabled the diversification of life on Earth by adding O₂ to the atmosphere.
- Human industrialization is intensifying the aerobic challenges to marine ecosystems.
- Historical observations report an 2% decline in upper-ocean O₂.
- The ocean retains under 2% of global O₂ inventory, while 98% stays in the atmosphere.

- The constraining role of O_2 for ocean species arises from two kinematic factors: lower solubility of gases in water and slower circulation of the ocean than of atmosphere.
- Industrialization of human societies produced two negative consequences for the ocean's O_2 reservoir: (1) nutrients mobilized for agriculture are carried to coastal oceans, stimulating phytoplankton growth and thus higher O_2 respiratory demand (2) global warming causes O_2 depletion throughout the ocean
- Coastal regions make up only 7% of ocean's area but contain 15% of it's net primary productivity and 90% of it's animal life.
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4.1 Glossary

4.1.1 O_2 solubility in seawater (K_H)

O_2 in seawater when in equilibrium with the O_2 in a gas phase, as a function of temperature, pressure, and salinity

4.1.2 O_2 partial pressure (pO_2)

the ratio of O_2 concentration to O_2 solubility

4.1.3 hypoxia

the condition wherein O_2 is insufficient to meet metabolic demand, which varies with species and temperature, in contrast to fixed definition commonly used in oceanography ($O_2 \downarrow 60\mu M$)

4.1.4 Eutrophication

increased growth, primary production, and biomass of algae in response to nutrient enrichment in the environment (e.g. sewage, fertilizer, and detergents)

4.1.5 Metabolic storm

an anomalous ocean weather event in which high temperatures cause elevated organismal O_2 demand simultaneous with decline in O_2 available in seawater

4.1.6 Temperature-dependent hypoxia

the threshold for hypoxia and its variation with temperature

4.1.7 Oxygen minimum zone (OMZ)

a vertical minimum in a water column; OMZs are globally prevalent but are strongest in the tropical thermoclines of the Indian and Pacific basins, where O_2 reaches traditionally defined hypoxic concentrations

4.1.8 Suboxia

$O_2 \downarrow 5 \mu M$, corresponding to the appearance of nitrogen deficits relative to phosphorous in seawater and the activation of anaerobic microbial metabolisms; used interchangeably with anoxia

4.1.9 Apparent O_2 utilization (AOU)

the amount of O_2 consumed by respiration over the transport time of a water parcel from the surface ocean to the interior

4.1.10 Critical O_2 pressure in seawater

the ambient pO_2 when an organism's O_2 supply balances its resting metabolic O_2 demand

5 Lecture 4: On measuring biodiversity and its importance

Lars Gamfeldt

5.1 Seminar overview

- What is biodiversity and how do we measure it?
- How do we know that biodiversity is declining, what metrics do we use?
- Why is it important to measure biodiversity change?

Biomass of different groups:

- Plants: 450 Gt C
 - Anthropods (e.g. insects): 1Gt C
 - Fish: 0.7Gt C
 - Wild mammals: 0.007 Gt C
 - Humans: 0.06 Gt C
 - Livestock: 0.1 Gt C
 - Wild birds: 0.002 Gt C
 - Annelids: 0.2 Gt C
- Animals: 2 Gt C
- Fungi: 12 Gt C
- Bacteria: 70 Gt C
- Protists: 4 Gt C
- Viruses: 0.2 Gt C
- 7 Gt C

5.2 What is biodiversity

ecosystems $\leftrightarrow$ species $\leftrightarrow$ genes

- Biodiversity is a concept not a quantity
- Typically, we recognise different dimensions of biodiversity
- We operationalise it in different ways

5.3 Biodiversity and human health: An analogy

physical, social and mental healths all overlap with each other

analogously, biodiversity has 4 different dimensions:

- evolutionary: gene diversity, phylogenetics etc.

- functional: trait diversity, trophic diversity etc.
- taxonomic: species diversity etc.
- abundance: population size, biomass etc.
- biodiversity consists of different dimensions
- use a variety of metrics to try and quantify these dimensions
- understanding biodiversity of a place, time etc.

5.4 Is biodiversity higher on land or in the sea?

Number of species on land is about 8M and about 2M in the ocean (both have huge margin of error).

77% of all animal species occur on land, 11% in freshwater, 12% are marine. Explanation: 90% of species of land are anthropods, most of which are insects.

5.5 Terrestrial Anthropods are extremely diverse

Erwin (1982) surveyed insects on 19 different trees of the same species, he found 1200 insect species.

There could be 1.5M species of beetles alone!

5.6 Is biodiversity higher on land or in the sea?

- It depends on how it is measured
- Depends on the value put on different dimensions of biodiversity

5.7 The Living Planet Index (WWF)

It reflects the average size of wildlife populations.

5.8 The Red List Index

Consists of 9 categories from Extinct to Least Concern (through Threatened).

5.9 Global extinction rates

Current extinctions are well above the background rate (expected rate of extinction).

5.10 Habitat loss: Deforestation

30% of forests remaining (1700 baseline)

70% of forests are within 1km of some human settlement or activity.

5.11 Why is it important to understand and measure biodiversity?

- Ethical reasons
- Functional reasons
- Nature's contribution to people (ecosystem services)

6 Lecture 6: Aquatic pollution

6.1 Definitions

Pollution

- Eutrophication
- Organic matter
- Rubbish
- Microbial
- **Hazardous substances / contaminants / toxicants**

6.2 Naturally occurring hazardous substances

- Heavy metals, PAHs, petroleum, hormones, ...
- essential and non-essential substances

Toxins – substances produced by living organisms that are toxic to humans.

6.3 Man-made hazardous substances

- 350000 registered
- 100000 on the market in Europe

6.4 Planetary boundaries

2009

- 7 boundaries assessed
- 3 crossed

2015

- 7 boundaries assessed
- 4 crossed

2023

- 9 boundaries assessed
- 6 crossed

6.5 Itai-itai (ouch-ouch) disease 1912

- cause: Cadmium (Cd) – heavy metal
- in Japan
- from a mine
- the cadmium went into the rivers, and the rivers were used to water the rice plants

6.6 Minamata disease 1956

- also Japan
- Mercury (Hg), also heavy metal
- Methyl mercury, a form of mercury transformed by bacteria
- again, originating from a factory

6.7 Effects on wild-life 1960s

- A decline in eagles in Sweden and north of Baltic Sea
- Due to DDT toxicants
- Measured in DDT toxic equivalents
- Clear correlation between DDT toxic equivalents and egg shell thickness

6.8 Biomagnification

- Concentration of DDT increases up the food chain

6.9 Tri-butyl-tin (TBT)

- Oysters exposed to TBT have different shapes and become sterile
- The shape is very degenerate of course

6.10 Oil spills

6.11 Sources, fate and status

- Input from diffuse and point sources
- Air/water exchange of Hz^2
- Wet + dry deposition of Hz
- Direct + indirect photolysis UV
- Chemical and biological transformation
- Sedimentation
- Chemical and biological transforming in the sedimentation

6.12 Hazardous substances are everywhere

6.13 Sewage treatment plants

Pain killers

- Diklofenak
- Naproxen
- Ketoprofen
- Ibuprofen

PFAS

- Recently contaminated drinking water in Skåne

6.14 Assessing status – Environmental quality standards

- Identify receptors and compartments at risk
- Collate and quality assess data
- Extrapolation
- Propose EQS
- Implement EQS

²hazardous substance

6.15 Chemical risk – fresh water

Acute exposure is not so bad in Europe but chronic exposure is pretty bad. Some regions have over 75% chemical risk.

6.16 Environmental impact – ellpout larvea deformities

6.17 Environmental impact – mussel hemolymph

6.18 Bottom-up – Single species tests

- Standard test species and protocols
- Low biodiversity
- High repeatability and reproducibility

6.19 Biodiversity – why it matters

- Impact of sewage in Greenland

6.20 Assessing reproductive success

6.21 Biodiversity and community ecotoxicology

Community: an assemblage or associations of populations of two or more different species, belonging to the same group, that occupy the same location

6.22 Top down – a more ecological approach

- Secondary producers (SP)
- Primary producers (PP)

6.23 Scrubbers – closed loop

7 Lecture 6: plastic contamination

- 350000 chemicals registered for production globally
- over 16000 chemicals used in plastics
- 10726 chemicals without hazard data
- 4219 chemicals classified as hazardous
- 1192 chemicals less hazardous

7.1 Impacts on biodiversity due to debris abundance

- Ingestion / entanglement
 - Entanglement: starvation, drowning, limits foraging. Mostly due to "ghost fishing", but not only
 - Ingestion: may lead to gut clogging, starvation. Accidental or confusion with prey
- Habitat destruction
 - Landfill, dumping
 - Flooding

- Smothering:
 - * mangrove forests: weight and shading effect affects sensitive vegetation

7.2 Impacts on biodiversity due to plastic-associated chemicals

7.3 Impacts of microplastics on biodiversity